



Simple Liquid Dispenser For Bottling Plants

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Automated liquid filler machines are used to pack various liquid items such as oils, milk, and beverages in their bottling plants. But these machines are costly and not affordable by most small bottlers. This simple liquid dispenser releases a litre of the liquid each time the button is pressed.

Block diagram of the dispenser shown in Fig. 1 illustrates its main stages.

Circuit and working

Circuit diagram of the simple liquid dispenser shown in Fig. 2 includes 5V, 7805 voltage regulator IC1, ATtiny85A AVR microcontroller (MCU) IC2, YF-S201 liquid flow sensor, common-cathode bi-colour LED BICO1, normally-closed solenoid valve, and BD139 NPN transistor T1.

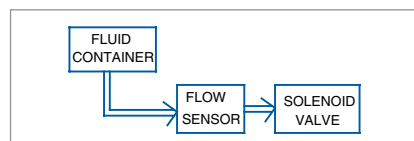


Fig. 1: Block diagram of the liquid dispenser

ATtiny85A AVR microcontroller is the brain of the circuit that controls the predefined quantity of the liquid flowing out. The flow sensor is connected to connector J1 and the solenoid valve is connected to connector J2.

In standby mode, the bi-colour LED emits red colour and the valve is de-energised. Every time button S1 is pressed, the valve gets energised and the liquid releases from the solenoid valve. The bi-colour LED blinks in green colour to indicate that the liquid filling is in process.

When liquid flow reaches the predefined quantity, the valve is de-energised by the MCU and now the bi-colour LED emits red colour to indicate that the filling process has been completed.

This cycle repeats whenever the button is pressed. Button S2 is used to manually reset the MCU.

Flow sensor

The flow sensor has a rotor along with a Hall effect sensor in the line of liquid passage. When liquid flows through the sensor, the rotor rotates. This

causes the Hall effect sensor to output a pulse. Thus, the rate of flow (Q) can be measured as follows:

Rate of flow (Q) in litres/min = Output pulse frequency/7.5

But we want the quantity in litres and not the flow rate in litres per minute. By re-arranging the above equation for litres, we get:

Number of litres = Number of output pulses/450

That means, it takes 450 pulses to flow one litre of liquid through the flow sensor.

Connections between ATtiny85 and Arduino Board during Programming

ATtiny85 pins	Arduino Uno board pins
Vcc (pin 8)	5V
GND (pin 4)	GND
Pin 7	13
Pin 6	12
Pin 5	11
Reset (pin 1)	10

PARTS LIST

Semiconductors:

- IC1 - 78L05, 5V voltage regulator
- IC2 - ATtiny85A microcontroller
- T1 - BD139 NPN transistor
- D1 - 1N4007 rectifier diode
- BICO1 - 3-pin common-cathode bi-colour LED

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 1.5-kilo-ohm
- R3 - 360-ohm
- R2 - 470-ohm
- R4 - 10-kilo-ohm

Capacitors:

- C1 - 0.33µF ceramic
- C2 - 0.01µF ceramic

Miscellaneous:

- CON1 - 2-pin terminal connector
- J1 - 3-pin connector
- J2 - 2-pin terminal connector
- L1 - 12V normally-closed solenoid valve
- S1, S2 - Tactile push switch
- F1 - YF-S201 water flow sensor
- 12V power supply

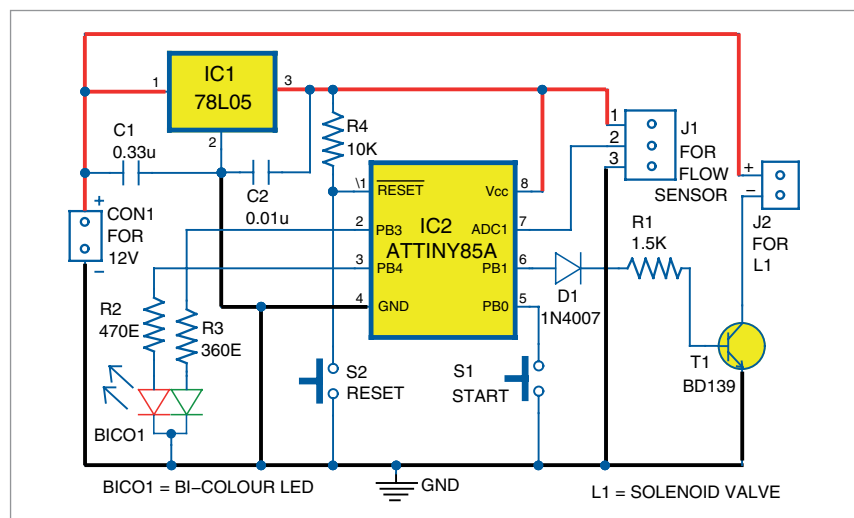


Fig. 2: Circuit diagram of the dispenser